Skills Summary

Identities, Multiple Angles, and Exact Solutions

Use this reference while completing your worksheet or when studying.

Which job is it? No equation to solve ⇒ simplify. A doubled angle ⇒ convert it. An interval given ⇒ produce a complete exact solution set.

1. Simplifying with the basic identities

Pythagorean: $\sin^2\theta + \cos^2\theta = 1$, $1 + \tan^2\theta = \sec^2\theta$, $1 + \cot^2\theta = \csc^2\theta$ (each also rearranges, e.g. $1 - \cos^2\theta = \sin^2\theta$).

Quotient / reciprocal: $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$, $\sec\theta = \dfrac{1}{\cos\theta}$, $\csc\theta = \dfrac{1}{\sin\theta}$.

Order of moves: factor a common term first; look for a Pythagorean form; only then rewrite everything in sines and cosines. Conjugate denominators multiply to a difference of squares, which the Pythagorean identity always collapses.

Example: Simplify $\dfrac{\sec^2 x - 1}{\sec^2 x}$.

Step 1. The numerator is exactly one side of a rearranged Pythagorean identity, so substitute instead of expanding into sines and cosines straight away.

Step 2. $\sec^2 x - 1 = \tan^2 x$, so the fraction is $\dfrac{\tan^2 x}{\sec^2 x} = \dfrac{\sin^2 x}{\cos^2 x} \cdot \cos^2 x$. $= \boldsymbol{\sin^2 x}$

Try it: Simplify $(1 - \cos^2 x)\csc x$.

check: $\boldsymbol{\sin x}$

2. Double-angle formulas: choosing the right form

$$\sin 2\theta = 2\sin\theta\cos\theta \qquad \tan 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta}$$

$$\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$$

The cosine has three forms on purpose. Pick the one written in the function you already know, or the one that makes a neighbouring term collapse ($1 + \cos 2\theta = 2\cos^2\theta$ and $1 - \cos 2\theta = 2\sin^2\theta$).

Never write $\sin 2\theta$ as $2\sin\theta$ — that is not an identity, and it can even leave the range of sine.

Example: $\sin\theta = \dfrac{5}{13}$ with θ in quadrant I. Find $\cos 2\theta$.

Step 1. Only the sine is known, so choose the cosine form written purely in sine and no quadrant reasoning is needed — the square removes the sign.

Step 2. $\cos 2\theta = 1 - 2\sin^2\theta = 1 - 2\left(\dfrac{5}{13}\right)^2 = 1 - \dfrac{50}{169}$. $\cos 2\theta = \boldsymbol{\dfrac{119}{169}}$

Try it: $\cos\theta = \frac{3}{5}$. Find $\cos 2\theta$.

check: $-\frac{7}{25}$

3. Exact solution sets on an interval

The routine:

- **One function.** Use an identity to remove every other trigonometric function (and every multiple angle).
- **Factor, never divide.** Dividing by $\sin x$ throws away the solutions where $\sin x = 0$.
- **Reference angle, then quadrants.** Each factor gives a value; take its reference angle and place it in both quadrants where the sign is right.
- **Sweep the interval.** On $[0, 2\pi)$ list every angle, including 0 itself, and keep them exact.

Example: Solve $2\cos^2 x + \cos x - 1 = 0$ on $[0, 2\pi)$.
Step 1. One function appears but squared, so this is a quadratic in cosine — factor it rather than isolating anything.
Step 2. $(2\cos x - 1)(\cos x + 1) = 0$, so $\cos x = \frac{1}{2}$ (reference angle $\frac{\pi}{3}$, quadrants I and IV) or $\cos x = -1$. $\boldsymbol{x = \frac{\pi}{3}, \pi, \frac{5\pi}{3}}$

Try it: Solve $2\sin^2 x + \sin x - 1 = 0$ on $[0, 2\pi)$.

check: $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

(!) Watch out: an interval question is not finished when you have one angle — each factor usually contributes two, and 0 counts. And check that every solution really lies in $[0, 2\pi)$: the right endpoint is excluded.